Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 10 [2018 P1]

Test Date: 07th May 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2018-P1-Model

Time: 3:00Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 7 – 14)	Questions with Numerical Value Type (e.g. 6.25, 7.00, 0.33, .30, 30.27, 127.30)	+3	0	8	24
Sec – III(Q.N : 15-18)	Paragraph Questions with Single Answer Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N : 25 –32)	Questions with Numerical Value Type (e.g. 6.25, 7.00, 0.33, .30, 30.27, 127.30)	+3	0	8	24
Sec – III(Q.N : 33-36)	Paragraph Questions with Single Answer Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				**18**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N:37 – 42)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II(Q.N :43–50)	Questions with Numerical Value Type (e.g. 6.25, 7.00, 0.33, .30, 30.27, 127.30)	+3	0	8	24
Sec –III(Q.N : 51-54)	Paragraph Questions with Single Answer Type (2 Comprehensions – 2 +2 = 4Q)	+3	-1	4	12
Total				**18**	**60**

PHYSICS

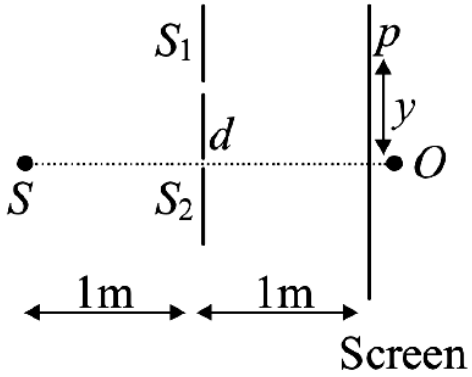
Max Marks:60

SECTION 1
(MULTIPLE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.
Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.

For Example: If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option) ,without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In an arrangement shown in the figure, d < < D, where d is the distance of separation of the slits S_1 and S_2 and $D = 1\,m$ is the distance between the slits and the screen. Source S is emitting monochromatic light of wavelength λ. Then



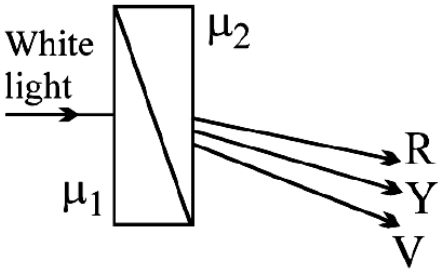
A) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{2}}$

B) Minimum value of d for the dark fringe at O is $\sqrt{\frac{\lambda}{3}}$

C) Corresponding to minimum value of d, the fringe width is $\sqrt{2\lambda}$

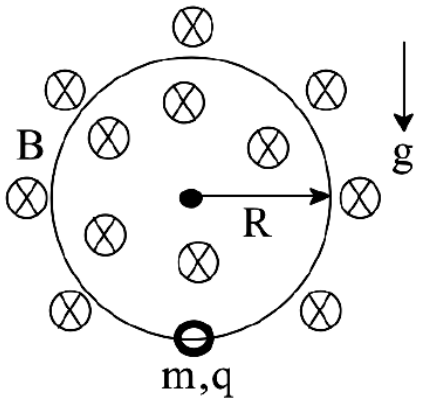
D) Corresponding to minimum value of d, the fringe width is $\sqrt{\frac{\lambda}{2}}$

2. A beam of white light is incident on a combinations of thin prisms with equal angles. In the out going beam the rainbow colours are observed as shown. μ stands for refractive index and θ for angle of dispersion. Then



A) $\mu_1 < \mu_2$ **B)** $\mu_2 = \mu_1$ **C)** $\theta_1 > \theta_2$ **D)** $\mu_1 > \mu_2$

3. The diagram shows a circular non–conducting smooth fixed track kept concentric to a magnetic field of magnitude B which exists in a cylindrical region of radius r(r > R). There is small bead of mass m and charge q rests on the track. The bead can move freely along track. If the magnetic field is switched off then



A) Impulse imparted to bead is $\frac{qRB}{2}$

B) Impulse imparted to bead is qRB

C) If $q=\frac{4m}{B}\sqrt{\frac{g}{R}}$ then bead will complete vertical circular motion

D) If $q=\frac{m}{B}\sqrt{\frac{g}{R}}$ then bead will complete vertical circular motion

4. Nucleus A decays to B with decay constant λ_1 and B decays to C with a decay constant λ_2. Initially at t = 0 number of nuclei of A and B are $2N_0$ and N_0 respectively. At $t=t_0$, number of nuclei B stop changing. If at this instant number of nuclei of B are $\frac{3N_0}{2}$, then

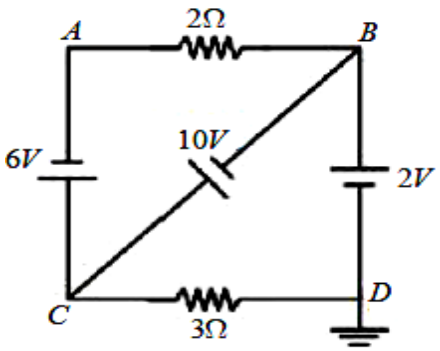
A) $t_0=\frac{1}{\lambda_1}\ell n\frac{4\lambda_1}{3\lambda_2}$

B) $t_0=\frac{1}{\lambda_2}\ell n\frac{4\lambda_1}{3\lambda_2}$

C) $N_A=\frac{3N_0}{2}\frac{\lambda_2}{\lambda_1}$ at $t=t_0$

D) $N_A=\frac{2N_0}{3}\frac{\lambda_2}{\lambda_1}$ at $t=t_0$

5. In the shown circuit:



A) Current passing through 2Ω resistance is 2 A

B) Current passing through 3Ω resistance is 4 A

C) Current in wire D to earth is zero

D) Potential of point A is 10 V

6. A man of mass m walks from end A to the other end B of a boat of mass M and length ℓ. The coefficient of friction between man and boat is μ and neglect any resistive force between boat and water. Choose the **CORRECT** statements?

A) If the man runs at his maximum acceleration the acceleration of the boat is $\frac{m}{M}\mu g$

B) Minimum time taken by man to reach other end of the boat is $\sqrt{\frac{2M\ell}{(M+m)\mu g}}$

C) Magnitude of displacement of center of mass of boat is $\frac{M\ell}{M+m}$

D) Velocity of centre of mass of man and boat system is zero

SECTION 2

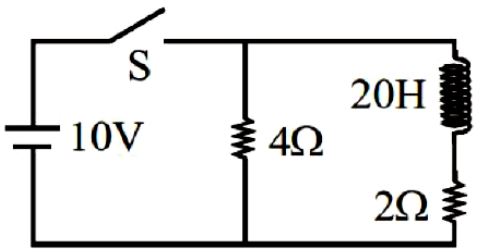
(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
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Zero Marks : 0 In all other cases.

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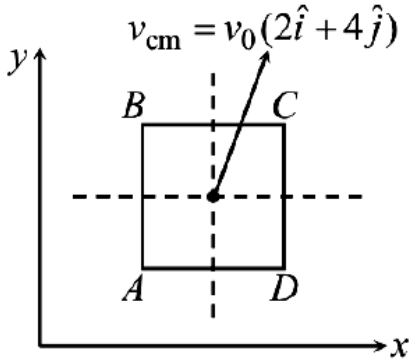
7. Temperature of a gaseous medium increases linearly from T to 2T over a distance x = 0 to x = ℓ, where molecular weight of gas is M and its adiabatic constant is γ. Time taken by sound to travel from x = 0 to ℓ is $k\sqrt{\frac{M\ell^2}{\gamma RT}}$ where k is _____ (take $\sqrt{2}=1.41$)

8. The switch S is closed at time t = 0. Find the time in seconds when the currents in the two resistances become equal.

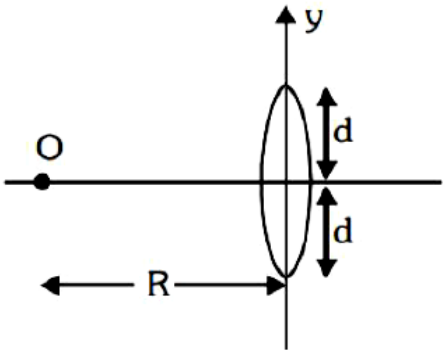


9. The vernier of a vernier scale is divided into 10 divisions which coincide with 9 divisions of the main scale, each main scale division being 0.5 mm. When the two jaws of the instrument are in contact with each other, the 4th division of the vernier scale coincides with a main scale division and the zero of the vernier lies to the right of the zero of the main scale. When a sphere is inserted between the jaws, the zero of vernier scale lies slightly to the right of 1.8 cm and the sixth vernier division coincides with a main division. The diameter of sphere will be (in cm)

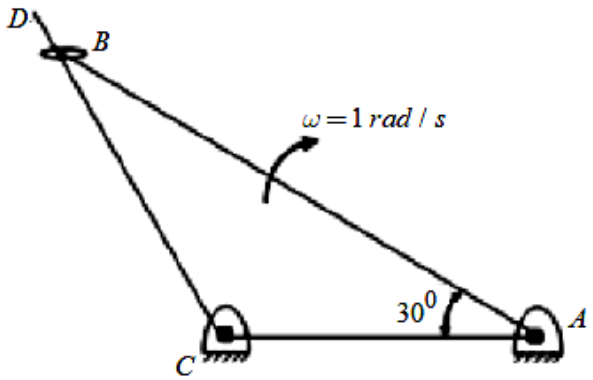
10. A uniform solid square plate $ABCD$ of mass m and side a is moving in $x-y$ horizontal smooth plane. The velocity of centre of mass is $v_0\left(2\hat{i}+4\hat{j}\right)m/s$. The end A of square plate is suddenly fixed by a pin, the new velocity of centre of mass of square is given by $v_0\left(a\hat{i}+b\hat{j}\right)$, find $b-a$.



11. A biconvex lens of radius of curvature R is made up of variable refractive index $\mu=2\left(1+\frac{|y|}{d}\right)$. Assume $2d<<R$. A point object is placed at $R=1\,m$ on the principal axis as shown in figure. If the spread of image lies over a span of n meter along the axis, then find the value of n.

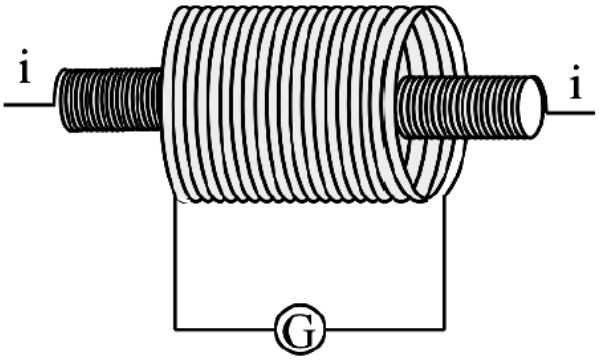


12. In the figure shown a smooth ring is connected to rod AB, while rod CD passes through ring. At the given instant angular velocity of rod AB about hinge A is $1\,rad/s$ and $AC=CB$, then instantaneous angular velocity of rod CD about hinge C is (in rad/s)

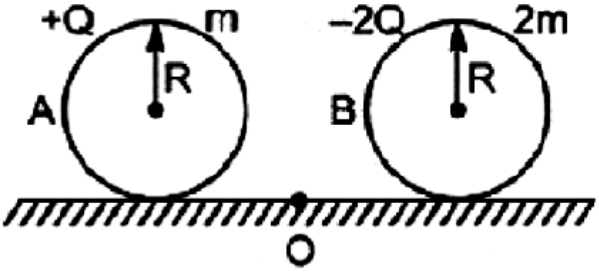


13. A long solenoid of radius $2\ cm$ has 100 turns/cm and is surrounded by a 100–turn coil of radius $4\ cm$ having a total resistance of $20\ \Omega$. The coil is connected to a galvanometer as shown in figure. If the current in the solenoid is changed from 5A in one direction to 5A in the opposite direction. The charge which flows through the galvanometer is $n\ \mu C$.

Then find value of n [Take $\pi^2 = 10$]



14. A hollow non–conducting sphere A and a solid non–conducting sphere B of equal radius 'R' and masses m and 2m are kept at a large distance apart on a rough horizontal surface. Charge on the two spheres A and B are Q and –2Q respectively. Charges are distributed uniformly (on surface in A and in volume in B) and remain constant and uniform as the spheres come closer. Friction is sufficient to support pure rolling and the kinetic energy of the two spheres just before collision is K_A and K_B. Find $100 \times \frac{K_A}{K_B}$.



SECTION 3
PARAGRAPHS

This section contains **TWO (02)** paragraphs. Based on each paragraph, there are **TWO (02)** questions.
Each question has **FOUR** options. **ONLY ONE** of these four options corresponds to the correct answer.
For each question, choose the option corresponding to the correct answer.
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PARAGRAPH "X"

(*There are two questions based on PARAGRAPH* "X", *the question given below is one of them*)

A copper wire of negligible mass, 1m length and cross–sectional area $10^{-6} m^2$ is kept on a smooth horizontal table with one end fixed. A ball of mass 1kg is attached to the other

end. The wire and the ball are rotating with an angular velocity of 20 rad/s. If the elongation in the wire is $10^{-3} m$.

15. The Young's modulus of the wire is

A) $4.0\times10^{11} N/m^2$ **B)** $2.5\times10^{11} N/m^2$

C) $2.0\times10^{11} N/m^2$ **D)** $4.5\times10^{11} N/m^2$

16. If for same wire as stated above the angular velocity is increase to 100 rad/s and the wire breaks down, the breaking stress is

A) $1.5\times10^{10} N/m^2$ **B)** $10^{10} N/m^2$

C) $2.0\times10^{10} N/m^2$ **D)** $3.0\times10^{10} N/m^2$

PARAGRAPH "A"

(*There are two questions based on PARAGRAPH* "A", *the question given below is one of them*)

A pair of stars rotate about its common centre of mass. One of the stars has a mass M and the other has mass m such $M = 2m$. The distance between the centres of the stars is d (d being large compared to the size of either star)

17. The period of rotation of the stars about their common centre of mass (in terms of d, m, G) is

A) $\sqrt{\frac{4\pi^2}{Gm}d^3}$ **B)** $\sqrt{\frac{8\pi^2}{Gm}d^3}$ **C)** $\sqrt{\frac{2\pi^2}{3Gm}d^3}$ **D)** $\sqrt{\frac{4\pi^2}{3Gm}d^3}$

18. The ratio of the angular momentum of the two stars about their common centre of mass $\left(L_m / L_M\right)$ is

A) 1 **B)** 2 **C)** 4 **D)** 9

CHEMISTRY

Max Marks: 60

SECTION 1

(MULTIPLE CORRECT ANSWER TYPE)

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correct options (either first or third or fourth option) ,without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

19. Certain nitrogen-containing compounds decompose thermally to release nitrogen gas (N_2). Which of the following statements is(are) correct?

A) Compounds containing the azide (N_3^-) ion decompose readily to produce N_2 gas.

B) All ammonium salts release N_2 gas upon heating.

C) Metal nitrides react with water to produce amide of corresponding metal.

D) Only organic nitrogen compounds can release N_2 gas on thermal decomposition.

20. Metal carbonyls are important organometallic compounds with unique bonding characteristics. Which of the following statements is(are) correct?

A) The bonding in metal carbonyls involves both σ-donation from the carbonyl ligand and π-backbonding from the metal to antibonding orbitals of CO

B) An increase in the oxidation state of the metal strengthens the metal–carbon bond due to increased electrostatic attraction.

C) Bond strength of carbonyl C–O bond increases when the oxidation state of the metal increases.

D) The 18-electron rule is commonly followed by stable metal carbonyl complexes.

21. In the Solvay process the following reactions takes place

i) $2NH_3 + H_2O + CO_2 \rightarrow (W)$

ii) $(W) + H_2O + CO_2 \rightarrow (X)$

iii) (X) + NaCl $\rightarrow$ M + N

iv) $(M) \xrightarrow{\Delta} P(s) + Q(v) + R(g)$

Which of the following statement(s) is/are correct?

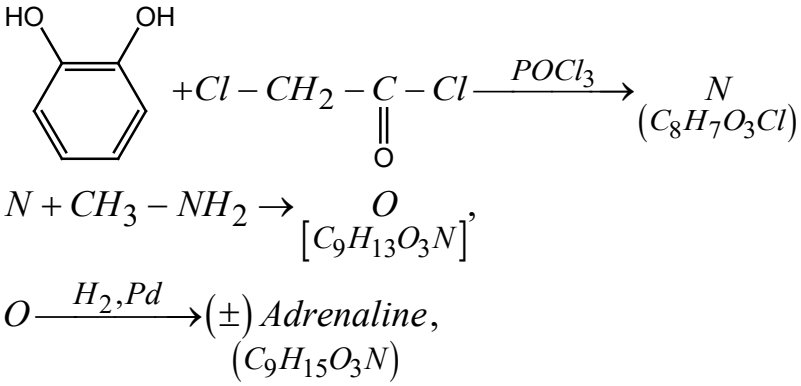
A) On adding colour less phenolphthalein indicator to aqueous solution of (M) there is no colour change

B) Aqueous solution of Compound (W) is more alkaline than (M)

C) Aqueous solution of Compound (P) is alkaline with hydronium ion concentration less than $10^{-7}\, mole\, lit^{-1}\, at\, 25°C$

D) Aqueous solution of (P) is less alkaline than (W), (X) or (M)

22. Adrenaline is prepared by using following sequence of reactions



Then, which of the following statements is/are correct?

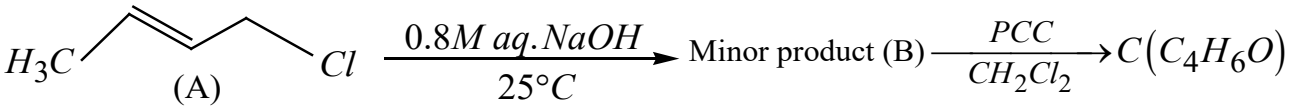
A) **N** on treatment with $NaOI$ followed by acidification produces 3,4-dihydroxy benzoic acid

B) Adrenaline contains amide linkage

C) Adrenaline on treating with neutral $FeCl_3$ produces violet colour

D) Adrenaline on treating with CH_2N_2 / BF_3 produces an ether.

23. Which of the option(s) is/are correct regarding the given sequence?



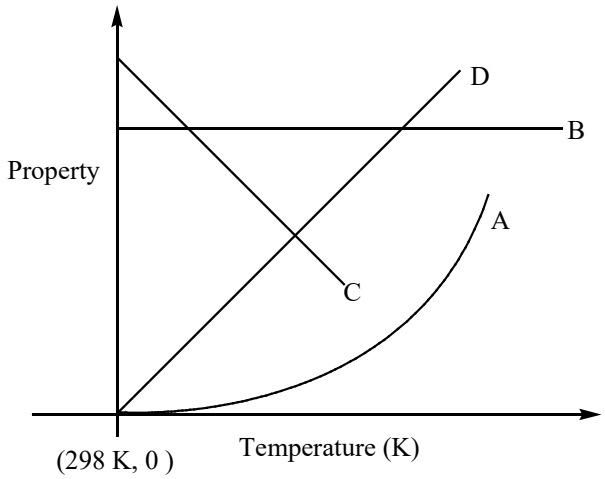
A) A to B is SN^1 reaction path only.

B) 'C' can be a conjugated ketone.

C) 'C' readily undergoes hydration than 1-Buetene across C=C

D) 'B' can give turbidity immediately with Lucas reagent.

24. The following graph represents the dependence of certain properties as function of temperature



Which of the following is correctly match the property with line/curve?

A) Curve A-The rate constant of a reaction with $E_a \neq 0\ kJ/mol$ and nonzero pre exponential factor

B) Line B – The molar heat capacity of an mono atomic ideal gas at constant volume

C) Line C – Solubility of gas in liquid solvent at constant pressure.

D) Line D – The osmotic pressure of ideal solution at a given concentration.

SECTION 2
(NUMERICAL VALUE TYPE)

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25. How many of the following compounds are paramagnetic in nature?

a) $\left[Co\left(C_2O_4\right)_3\right]^{3-}$ b) $\left[CoF_4\right]^{3-}$ c) $\left[FeF_6\right]^{3-}$ d) $\left[MnCl_6\right]^{3-}$

e) $\left[Fe\left(CN\right)_6\right]^{3-}$ f) $\left[Ni\left(CO\right)_4\right]$ g) O_3 h) O atom

i) Dioxygen molecule

Atomic number : Mn = 25, Fe = 26, Co = 27, Ni = 28

26. Urea reacts with sodium hydroxide to produce ammonia gas, which is then completely utilized by copper (II) sulfate pentahydrate $\left(CuSO_4.5H_2O\right)$ to form a stable coordination compound. Assume that both reactions proceed with 100% efficiency. If 1200 g of urea and 1249 g of $CuSO_4.5H_2O$ are used in the preparation, determine the total mass (in grams) of sodium carbonate and the copper-ammonia coordination compound formed.
Atomic weights (g/mol): H = 1, C = 12, N = 14, O = 16, S = 32, Na = 23, Cu = 63.5.

27. A hypothetical FCC alloy A_xB_y was found to have 40% corners and 60% face centres occupied by 'A' and 60% corners and 40% face centres occupied by 'B'. If 'x' and 'y' are integers, then find the minimum value of $|x-y|$.

28. A fuel cell operates using methane gas $\left(CH_4(g)\right)$ and produces carbonate ions $\left(CO_3^{2-}\right)$ at the anode. It is used to power a car with a current of 80 A for a duration of 0.965 hours. Determine the volume of $CH_4(g)$ required at STP, assuming 100% efficiency.

Given:

Molar volume at STP $\left(V_m\right) = 22.7L/mol$

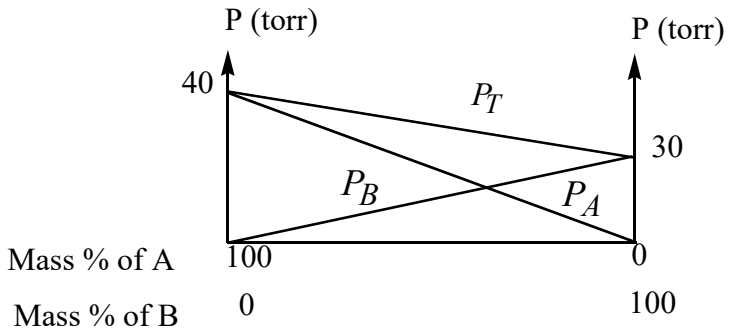
Faraday constant (F) =96500 C/mol

Report your answer in Litres

29. A definite amount of an ideal gas is taken from thermodynamic state – I to state – II isothermally at 300 K. If the process is carried out reversibly then work done is –1750 kJ and if process is carried out irreversibly then work done is –1000 kJ. Calculate ΔS_{total} (in kJ/K) in irreversible isothermal process.

30. If equal masses of A and B are present in a liquid solution then what is the total pressure exerted by vapours in torr above the liquid mixture?

Given:



Molecular masses of A and B are 80 and 120 respectively

31. To 1 litre solution containing 0.2 M Al^{+3} and 0.2 M Zn^{+2}, $NaOH$ is added slowly. If concentration of $\left[Al^{+3} \right]$ is 16×10^{-x} at the time when $Zn(OH)_2$ just begins to precipitate, then what is the value of x ?

$K_{sp}\left[Al(OH)_3 \right] = 16 \times 10^{-25}$ $\quad$ $K_{sp}\left[Zn(OH)_2 \right] = 2 \times 10^{-13}$

32. In a 2.24 L vessel, oxygen gas molecules were adsorbed on a solid surface of 100 cm^2 in area causing a decrease in partial pressure of oxygen from 1 atm to 0.5 atm at 273 K. The number of active sites on solid surface is $10^{24} m^{-2}$. How many oxygen molecules are adsorbed per active site?

Given,

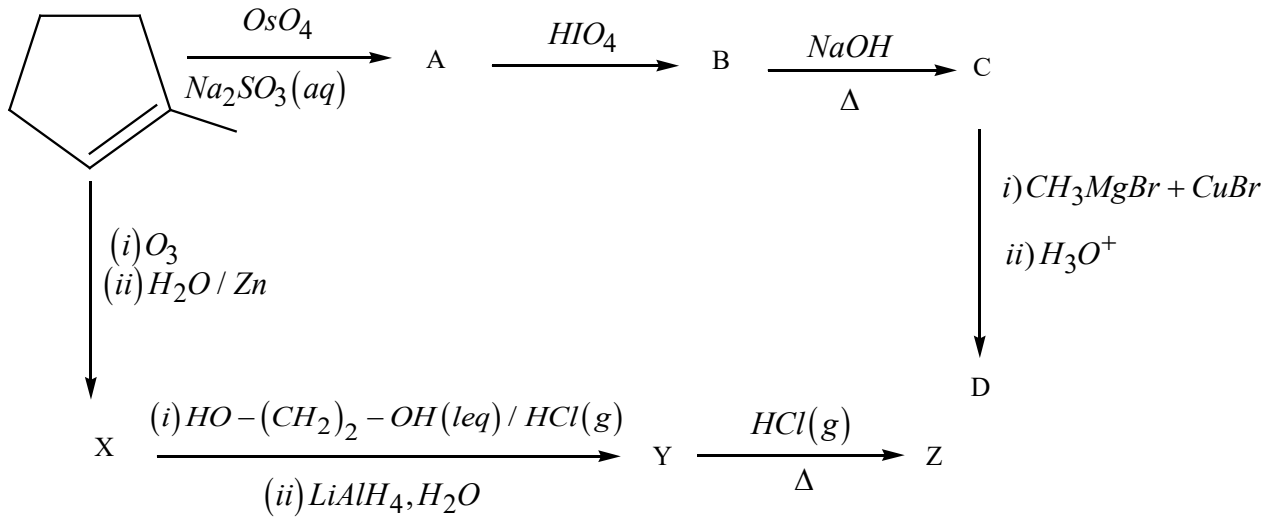
$R = 0.0821\ atm\ L\ mol^{-1} K^{-1}$

SECTION 3
PARAGRAPHS

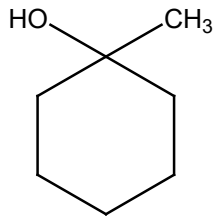
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Each question has **FOUR** options. **ONLY ONE** of these four options corresponds to the correct answer.
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Negative Marks : −1 In all other cases.

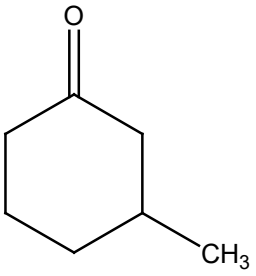
PARAGRAPH "X"

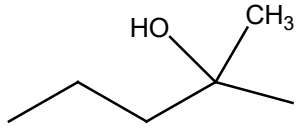
(*There are two questions based on PARAGRAPH* "X", *the question given below is one of them*)

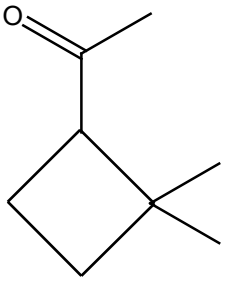


33. The product 'D' is

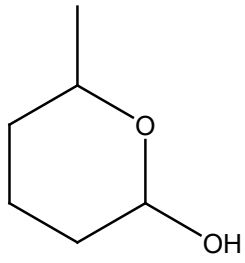
A) 

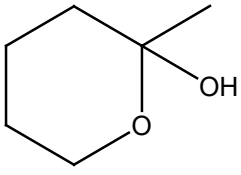
B) 

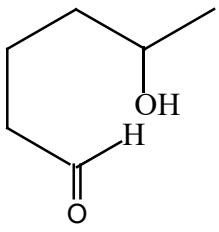
C) 

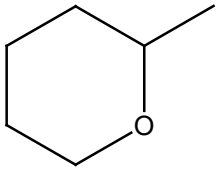
D) 

34. The prudct 'Z' is

A) 

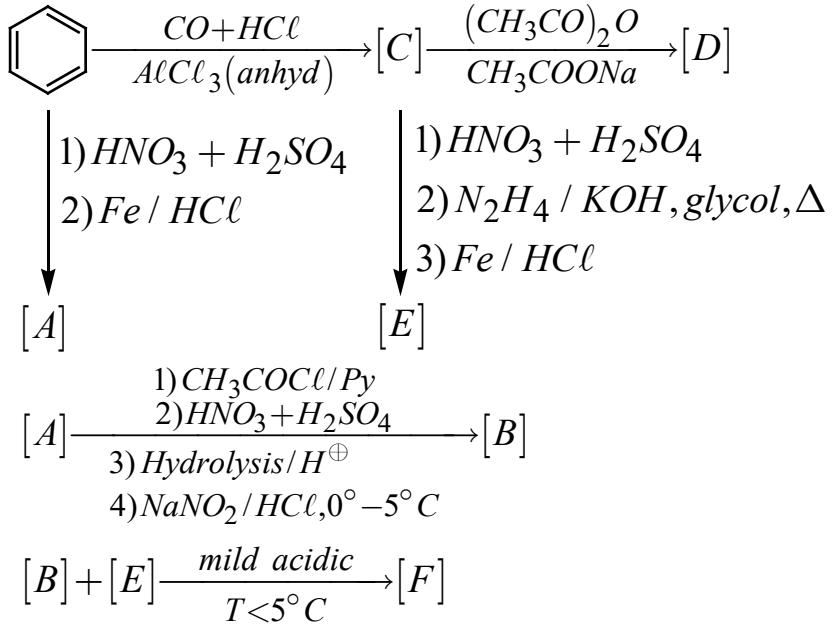
B) 

C) 

D) 

PARAGRAPH "A"

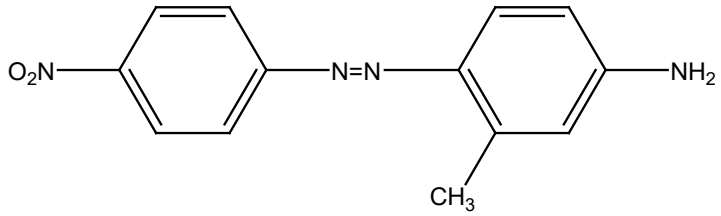
(*There are two questions based on PARAGRAPH* "A", *the question given below is one of them*)



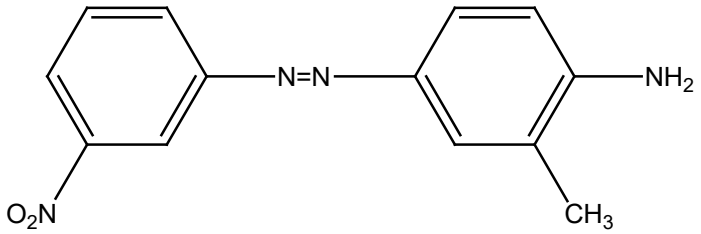
Consider major product in each reaction

35. Identify [F] product?

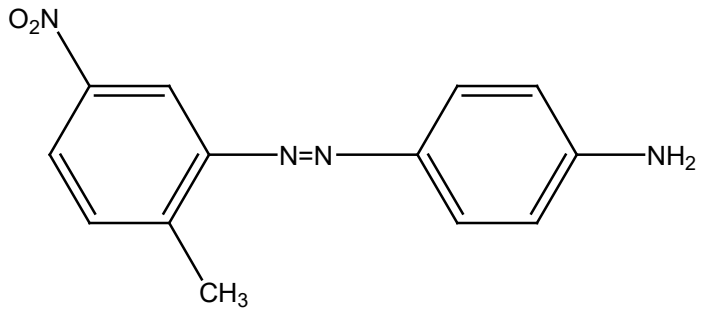
A)



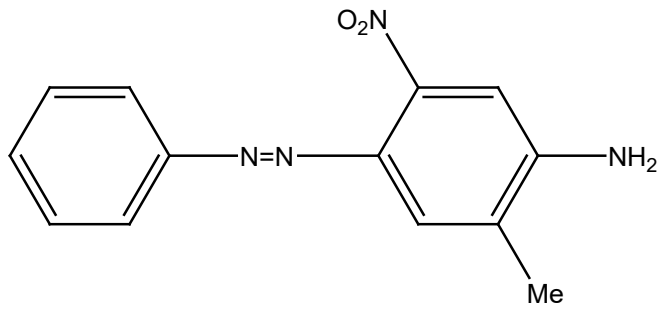
B)



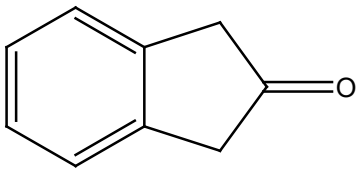
C)

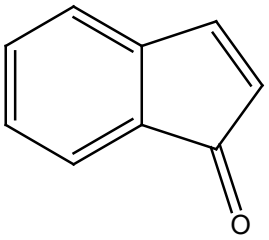


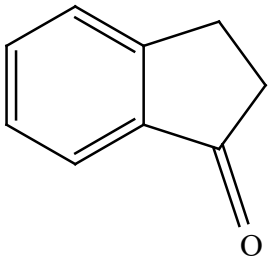
D)

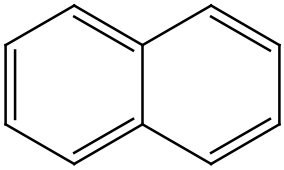


36. $[D] \xrightarrow[2)\,SOCl_2]{1)\,H_2/Ni} \xrightarrow[3)\,AlCl_3(anhydrous)]{}$ Major product is-

A) 

B) 

C) 

D) 

MATHEMATICS

Max Marks: 60

SECTION 1

(MULTIPLE CORRECT ANSWER TYPE)

This section contains **SIX (06)** questions.
Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
For Example: If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option) ,without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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37. Let $\vec{u}$ and $\vec{v}$ be unit vectors. If $\vec{w}$ is a vector such that $\vec{w}+\left(\vec{w}\times\vec{u}\right)=\vec{v}$, then

A) Maximum value of $\left[\vec{u}\,\vec{v}\,\vec{w}\right]$ is rational number.

B) Maximum value of $\left[\vec{u}\,\vec{v}\,\vec{w}\right]$ is irrational number.

C) When maximum of $\left[\vec{u}\,\vec{v}\,\vec{w}\right]$ occurs $\vec{u}$ and $\vec{v}$ are perpendicular.

D) When maximum of $\left[\vec{u}\,\vec{v}\,\vec{w}\right]$ occurs $\vec{w}$ and $\vec{u}$ are perpendicular.

38. If C_1 and C_2 are two concentric circles of radii 4 and 5 respectively with centre at origin. A tangent is drawn to circle C_2 at a given point 'P'. This tangent is the diameter of a variable circle C_3, which touches C_1 externally. Also this tangent meet C_3 in Q & R. If the diameter QR subtends a constant non-zero angle at a fixed point T on line joining O & P. Then the locus of the point 'T' for various positions of P on C_2 is

A) $x^2+y^2=2$ **B)** $x^2+y^2=4$ **C)** $x^2+y^2=64$ **D)** $x^2+y^2=36$

39. If $f(x), g(x)$ are thrice differentiable function on [0, 3] satisfying

$f'''(x) = g'''(x), f''\left(\frac{1}{2}\right) = g''\left(\frac{1}{2}\right) = 0, f'(1) = 4, g'(1) = 2$ and $f(3) = 12, g(3) = 3$, then

which of the following is (are) true?

A) $f(2) - g(2) = 7$

B) $|f(x) - g(x)| < 5 \Rightarrow x \in (-4,1)$

C) $|f(x) - g(x)| > 5 \Rightarrow x \in (-\infty,-4) \cup (1,\infty)$

D) $f(x) - g(x) = 2x - 3$

40. The curve(s) satisfying differential equation $y\left(\frac{dy}{dx}\right)^2 + (x-y)\frac{dy}{dx} - x = 0$ is/are:

A) Circle **B)** Straight line **C)** Parabola **D)** Ellipse

41. All the three roots of $az^3 + bz^2 + cz + d = 0$ have negative real parts $(a,b,c \in R, a \neq 0)$ then

A) $ab > 0$ **B)** $bc > 0$ **C)** $ad > 0$ **D)** $a > 0$

42. A set of lines $x + y - 2 + \lambda_1(2x + y - 3) = 0$ represents incident rays on an ellipse S = 0 and $2x + 3y - 23 + \lambda_2(2x - y - 3) = 0$ represents the set of reflective rays from the ellipse where $\lambda_1, \lambda_2 \in R$. If (3, 7) is a point on the ellipse normal at which meets the major axis at N. Then which of the following is/are **CORRECT**?

A) Eccentricity of ellipse is $\frac{\sqrt{5}}{2\sqrt{2}+1}$

B) N divides line segment joining two foci in the ratio $2\sqrt{2}:1$

C) Area of triangle formed by formed by point P and two foci 5

D) Eccentricity of ellipse is $\frac{\sqrt{5}}{2\sqrt{2}-1}$

SECTION 2

(NUMERICAL VALUE TYPE)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

43. A function is defined for all real numbers and satisfies $f(2+x)=f(2-x)$ and $f(7+x)=f(7-x)$ for all real x. If $f(0)=0$, what is the least number of roots $f(x)=0$ must have in the interval $-1000 \le x \le 1000$?

44. The sum of $\sin^2\frac{2\pi}{7}+\sin^2\frac{4\pi}{7}+\sin^2\frac{8\pi}{7}$ is S, then the value of 4S is_____.

45. A country has food deficit of 10%. Its population grows continuously at a rate of 3% per year. Its annual food production every year is 4% more than that of the last year. Assuming that the average food requirement per person remains constant, the country will becomes self-sufficient in food after n years, where n is the smallest integer bigger than or equal to $\frac{\log 10-\log a}{(\log b)-c}$ then the value of a + b + c is :

46. Let $\omega \neq 1$, be a 13th root of unity, find the remainder when $\prod_{k=1}^{12}\left(2-2\omega^k+\omega^{2k}\right)$ is divided by 1000.

47. A line $3x+y=8$ touches a hyperbola $H=0$ at $P(1, 5)$ and meets its asymptotes at A and B. If $AB=2\sqrt{10}$. If $C(1,1)$ be the centre of hyperbola, e and ℓ are eccentricity and latus rectum, (here length of conjugate axis < length of transverse axis), then value of $\left[e^2+\ell^2\right]$ is (Where [.] Greatest Integer Function)

48. In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. If the number of students who did not opt for any of the three courses is λ and α is number of positive integral factors of λ then number of equivalence relations defined from set A to set A where $n(A)=\alpha$ is :

49. The sum of an infinite geometric series whose first term is $\lim_{x\to 0}\sum_{K=1}^{2013}\frac{\left\{\frac{x}{\tan x}+2K\right\}}{2013}$ and whose common ratio is the value of $\lim_{x\to 0}\frac{e^{\tan^3 x}-e^{x^3}}{2\ln\left(1+x^3\sin^2 x\right)}$ is:

(where {.} is fractional part function)

50. If the number of three element subsets drawn from the set $S=\{1,2,3,.....,29,30\}$ are such that the sum of the three elements is a multiple of 3 is P, then the value of $\frac{P}{340}$ is:

SECTION 3
PARAGRAPHS

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PARAGRAPH "X"

(*There are two questions based on PARAGRAPH* "X", *the question given below is one of them*)

If $x,y,z\in R, x+y+z=4, x^2+y^2+z^2=6$. If the range of z is [p, q], then answer the following questions.

51. If $f(x)=\frac{q^2x\left(x^2+1\right)}{x^2+\left(x^2+1\right)^2}, x\geq 0$, then range of $f(x)$ is:

A) $[0,2]$ **B)** $\left[0,\frac{8}{5}\right]$ **C)** $\left[0,\frac{16}{5}\right]$ **D)** $\left[0,\frac{12}{5}\right]$

52. Number of integral values of a for which exactly one root of the equation $x^2+(a+1)x-4a=0$ lies in $[p,q]$ is:

A) 0 **B)** 2 **C)** 3 **D)** 4

PARAGRAPH "A"

(*There are two questions based on PARAGRAPH* "A", *the question given below is one of them*)

Let $x-y+1=0$ be axis of parabola, $x+y-4=0$ is tangent of same parabola at its vertex and $2x-y+3=0$ is one of its tangents. Then answer the following questions.

53. If the equation of directrix of parabola is $lx+my=29$, then $l+m$ is equal to :

A) 9 **B)** 13 **C)** 18 **D)** 23

54. If the length of latus rectum of parabola is $\frac{p\sqrt{2}}{q}$ where p and q are relatively prime natural numbers, then (p +q) is equal to:

A) 7 **B)** 11 **C)** 19 **D)** 23